On-shell Approaches for Gravitational Wave Physics

Waveform and beyond

Fei Teng (滕飞)

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YITP

Outline

Introduction: amplitudes meet GW

EFT matching

Spinning binaries from classical amplitudes

Observable based formalism (KMOC)

Outlook and conclusion

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Gravitational wave: new window to probe our Universe

GWTC-5: about 390 events

- Probe dynamics of black holes
- Test general relativity
- Probe new physics beyond GR
- Multimessenger astronomy/cosmology

Future ground based observatories

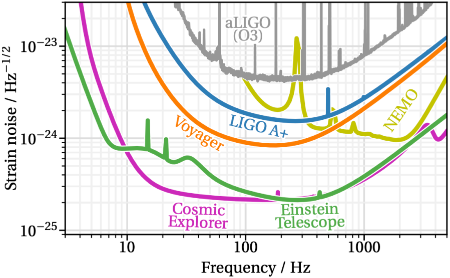
- Einstein Telescope
- Cosmic Explorer

Future space based observatories

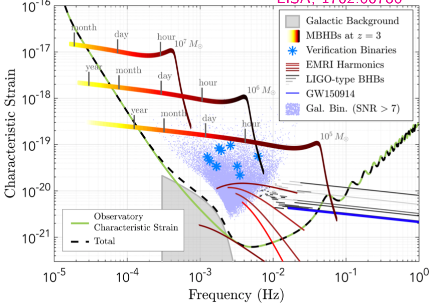
- LISA
- TaiJi
- TianQin

Require accurate theoretical prediction

cosmicexplorer.org

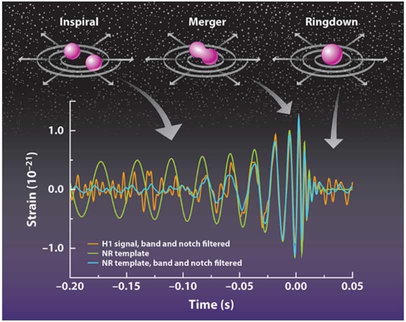


LISA, 1702.00786



GW from binary system

Numerical relativity provides the truth but computationally expensive



Accurate theoretical prediction of the GW production puts challenges on controling higher order perturbative expansions

$$H = \frac{\mathbf{p}_1^2}{2m_1} + \frac{\mathbf{p}_2^2}{2m_2} - \frac{Gm_1m_2}{|\mathbf{r}|} + \text{(corrections from general relativity)}$$

How to organize perturbations?

- Post-Newtonian (PN) expansion

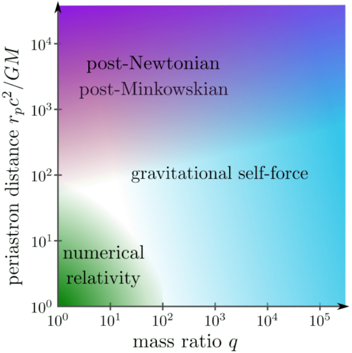
$$v^2 \sim \frac{Gm}{r} \ll 1$$

- Post-Minkowskian (PM) expansion

$$\frac{Gm}{r} \ll v^2 \sim 1$$

- Self-force expansion [van de Meent's talk]

$$\frac{Gm}{r} \sim v^2 \sim 1\,, \quad \frac{m_1}{m_2} \ll 1$$



Khalil, Buonanno, Steinhoff, Vines, 2204.05047

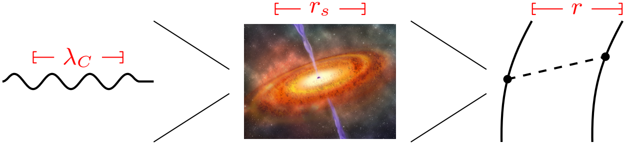
Field-theory based methods naturally lead to PM expansion

Why field theory and on-shell method?

- Efficient on-shell techniques: generalized unitarity, on-shell recursion, double copy, etc
- Powerful and versatile EFT construction: systematic handling of physics across scales
- Advanced integration techniques: IBP reduction, differential equation, etc

EFT perspective

From quantum to classical two-body problem



$\lambda_C \ll r_s \Longrightarrow Gm^2 \gg 1$

Classical contribution is encoded in every loop order

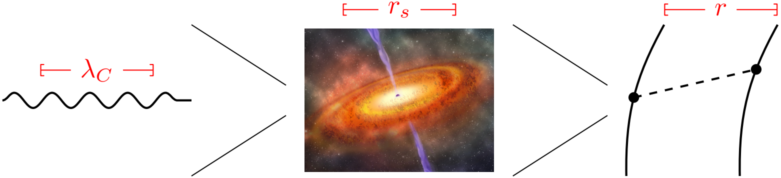
$r_s \ll r \Longrightarrow Gmq \ll 1$

Gmq is the small parameter in the classical perturbative expansion

$\lambda_C \ll r \Longrightarrow q/m \ll 1$ or $J \gg 1$ [also $S \gg 1$ in the presence of spin]

Keep only the leading order in the small momentum-transfer expansion
(Bohr's correspondence principle)

EFT perspective



Full theory: $S_{\text{full}} = -\dfrac{1}{16\pi G}\displaystyle\int d^4x\sqrt{-g}R - \dfrac{1}{2}\int d^4x \sum_{i=1,2}\left(g^{\mu\nu}\partial_\mu\phi_i\partial_\nu\phi_i - m_i^2\phi_i^2\right) + \mathcal{O}(R^2\phi^2)$

[graviton: $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$]

KMOC

Classical limit $(q, \ell, G) \to (\hbar q, \hbar \ell, \hbar^{-1} G)$

Integrate out soft gravitons

method of regions

Beneke, Smirnov, hep-ph/9711391

Observables

(classical)

Eikonal formula

$\mathcal{M}_{\text{QFT}} = \mathcal{M}_{\text{EFT}}$

EOM

Solve V_{PM} by matching amplitudes

V_{PM} given by ansatz

Effective theory: $S_{\text{eff}} = -\displaystyle\int dt \left[m_1\sqrt{1-\mathbf{v}_1^2} + m_2\sqrt{1-\mathbf{v}_2^2} + V_{\text{PM}}\right]$

Latest progress

	deflection & spin kick					waveform				
order	plain	spin-orbit	spin-spin	spin^2	tidal	plain	spin-orbit spin-spin	spin^2	tidal	Integration complexity
1PM	WQFT WEFT Amps HEFT	WQFT WEFT Amps HEFT	WQFT WEFT Amps HEFT	Amps HEFT	X	trivial	trivial	trivial	trivial	~ tree-level
2PM	WQFT WEFT Amps HEFT	WQFT WEFT Amps HEFT	WQFT WEFT Amps HEFT	Amps HEFT	WQFT WEFT Amps	WQFT WEFT Amps HEFT	WQFT WEFT Amps HEFT	Amps	WQFT WEFT Amps	~ 1-loop
3PM cons	WQFT WEFT Amps HEFT	WQFT Amps	WQFT Amps	Amps	WQFT WEFT	Amps HEFT	Amps			~ 2-loop
3PM diss	WQFT WEFT Amps HEFT	WQFT	WQFT		WQFT WEFT					
4PM cons	WQFT WEFT Amps	WQFT			WQFT					~ 3-loop
4PM diss	WQFT WEFT Amps	WQFT			WQFT					
5PM cons (2SF)	WQFT Amps									~ 4-loop

Updated from Jan Plefka's talk at MIAPbP 2024

WQFT: see Jakobsen, Mogull, Plefka, Steinhoff
WEFT: see Kälin, Porto
HEFT: see Brandhuber, Chen, Travaglini, Wen; Aoude, Haddad, Helset, Damgaard

Current frontier: partial results at **5PM-2SF**

Bern, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Zeng, 2509.17412
Bern, Herrmann, Roiban, Ruf, Smirnov, Smith, Zeng, 2512.23654
Driesse, Jakobsen, Mogull, Nega, Plefka, Sauer, Usovitsch, 2601.16256

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Classical potential and field theory

- Pioneering work by Iwasaki Yoichi (岩崎洋一)

Fourth-Order Gravitational Potential Based on Quantum Field Theory.

Y. Iwasaki

Research Institute for Fundamental Physics, Kyoto University - Kyoto

(ricevuto l'1 Marzo 1971)

Loop is classical!

(point out the mistake in Corinaldesi)

- Effective field theory: $\exp[S_{\text{eff}}] = \int \mathrm{D}\phi_{\text{short}} \exp[S_{\text{full}}]$

Goldberger, Rothstein, hep-th/0409156
Porto, gr-qc/0511061 Levi, Steinhoff, 1501.04956
and many more

- Classical interaction from amplitudes through "universality"

Vaidya, 1304.7475
Bjerrum-Bohr, Donoghue, Vanhove, 1309.0804
Bjerrum-Bohr, Donoghue, Holstein, Plante, Vanhove, 1410.7590

We want more synergy between EFT and on-shell amplitudes

EFT matching Cheung, Rothstein, Solon, 1808.02489

- Full theory: Schwarzschild black hole $\implies$ scalar field ϕ

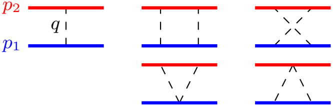
$$\mathcal{L} = \sqrt{-g}\left[-\frac{1}{16\pi G}R + \frac{1}{2}\sum_{i=1,2}\left(g^{\mu\nu}\partial_\mu\phi_i\partial_\nu\phi_i - m_i^2\phi_i^2\right)\right] + \mathcal{O}(R^2\phi^2)$$

- Effective theory: potential $V(\mathbf{k},\mathbf{k}')$ given by an ansatz

$$L = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\left[\sum_{i=1,2} a_i^\dagger(-\mathbf{k})\left(i\partial_t + \sqrt{\mathbf{k}^2 + m_i^2}\right)a_i(\mathbf{k}) - \int \frac{\mathrm{d}^3\mathbf{k}'}{(2\pi)^3} V(\mathbf{k},\mathbf{k}')a_1^\dagger(\mathbf{k}')a_1(\mathbf{k})a_2^\dagger(-\mathbf{k}')a_2(\mathbf{k})\right]$$

- Solve the EFT potential by matching the full theory and EFT amplitudes order-by-order in G in the classical limit

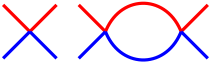
$$\mathcal{M}^{(L)} \xrightarrow{\text{classical limit}} \mathcal{M}^{(L)}_{\text{EFT}}$$



$q \to \hbar q$

$\ell \to \hbar \ell$

$G \to \hbar^{-1} G$



no massless loops (quantum)
no massive contact vertices (delta-function potential)

$$\mathcal{M}^{(L)}_{\text{classical}} \sim \frac{Gm^4}{\hbar^3 q^2}(Gmq)^L$$

$$E_1 = \sqrt{\mathbf{p}^2 + m_1^2} \qquad E_2 = \sqrt{\mathbf{p}^2 + m_2^2}$$
$$\gamma = \frac{E_1 + E_2}{m_1 + m_2} \qquad \xi = \frac{E_1 E_2}{(E_1 + E_2)^2}$$
$$\nu = \frac{m_1 m_2}{(m_1 + m_2)^2} \qquad \sigma = \frac{p_1 \cdot p_2}{m_1 m_2}$$

Hamiltonian: $H = E_1 + E_2 + \sum_{n=1}^{\infty} \frac{G^n}{|\mathbf{r}|^n} c_{n\mathrm{PM}}(\mathbf{p}^2)$

$$c_{1\mathrm{PM}} = -\frac{\nu^2 (m_1 + m_2)^2}{\gamma^2 \xi} (2\sigma^2 - 1)$$

Westpfahl and Goller 1979

$$c_{2\mathrm{PM}} = -\frac{\nu^2 (m_1 + m_2)^3}{\gamma^2 \xi} \left[\frac{3(5\sigma^2 - 1)}{4} - \frac{4\nu\sigma(2\sigma^2 - 1)}{\gamma\xi} + \frac{\nu^2 (1 - \xi)(2\sigma^2 - 1)^2}{2\gamma^3 \xi^2} \right]$$

Bel, Damour, Deruelle, Ibanez, Martin, 1981
Westpfahl 1985

$$c_{3\mathrm{PM}} = \frac{\nu^2 m^4}{\gamma^2 \xi} \left[\frac{3 - 6\nu + 206\nu\sigma - 54\sigma^2 + 108\nu\sigma^2 + 4\nu\sigma^3}{12} - \frac{4\nu(3 + 12\sigma^2 - 4\sigma^4) \operatorname{arcsinh} \sqrt{\frac{\sigma - 1}{2}}}{\sqrt{\sigma^2 - 1}} \right.$$
$$- \frac{3\nu\gamma(2\sigma^2 - 1)(5\sigma^2 - 1)}{2(1 + \gamma)(1 + \sigma)} + \frac{3\nu\sigma(20\sigma^2 - 7)}{2\gamma\xi} + \frac{\nu^2 (3 + 8\gamma - 3\xi - 15\sigma^2 - 80\gamma\sigma^2 + 15\xi\sigma^2)(2\sigma^2 - 1)}{4\gamma^3 \xi^2}$$
$$\left. + \frac{2\nu^3 (3 - 4\xi)\sigma(2\sigma^2 - 1)^2}{\gamma^4 \xi^3} - \frac{\nu^4 (1 - 2\xi)(2\sigma^2 - 1)^3}{2\gamma^6 \xi^4} \right]$$

Bern, Cheung, Roiban, Solon, Shen, Zeng, 1901.04424

State-of-the-art: partial results at $c_{5\mathrm{PM}}^{\mathrm{hyp}\ 2\mathrm{SF}}$

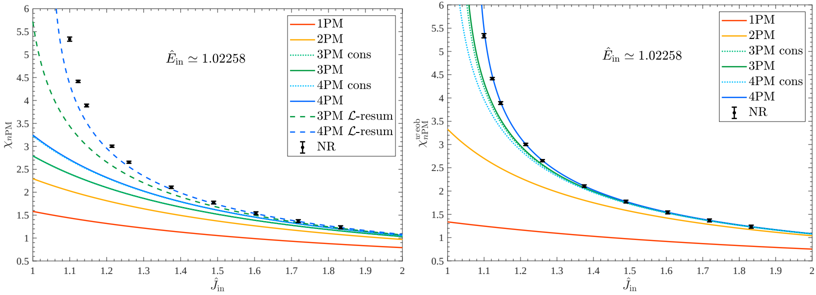
Bern, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Zeng, 2509.17412
Bern, Herrmann, Roiban, Ruf, Smirnov, Smith, Zeng, 2512.23654
Driesse, Jakobsen, Mogull, Nega, Plefka, Sauer, Usovitsch, 2601.16256

Comparison with numerical relativity [Damour's talk]

Scattering process

Effective-one-body (EOB) resummation [Buonanno, Damour, gr-qc/9811091] significantly improves the agreement with NR result!

Observable: scattering angle



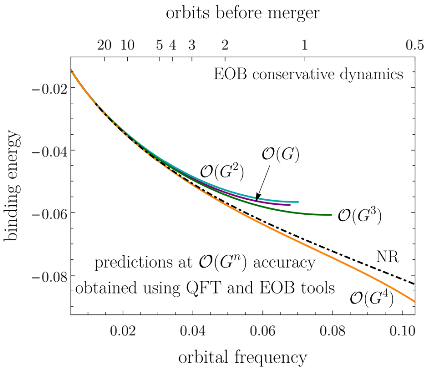
Damour, Rettegno, 2211.01399

Comparison with numerical relativity

Bound system

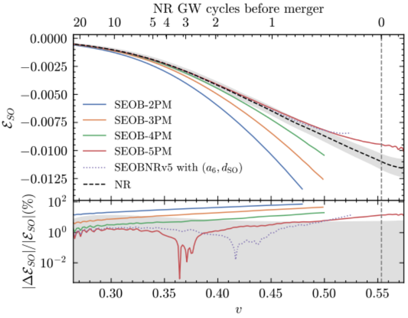
PM-informed two-body effective Hamiltonian

- Local contribution: direct analytic continuation
- Nonlocal (tail) contribution: supplement PN results from GR computations



Khalil, Buonanno, Steinhoff, Vines, 2205.05047

Snowmass white paper, 2204.05194



Buonanno, Mogull, Patil, Pompili, 2405.19181

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A survey [see Cangemi's talk for alternative amplitude-based approaches]

Stolen from Radu Roiban

Eight/Nine amplitudes-based approaches to higher-spin interactions

- 3-point "minimal" amplitude for arbitrary-spin particles — Arkani-Hamed, Huang, Huang building on Conde, Marzzola, Joung , Mkrtchyan
  - Naïve higher-point amplitudes with spurious poles at $S^{\geq 5}$; fixed (2022) Chen, Chung, Huang, Kim
- Exponentiated soft factors — Guevara, Ochirov, Vines
  - Same higher-point spurious pole issue as AHH at $S^{\geq 5}$
- Heavy-particle EFT-like theory — Aoude, Haddad, Helset
- Worldline QFT with spin — Jakobsen, Mogull, Plefka, Steinhoff; Haddad, Jakobsen, Mogull, Plefka
- Fixed order in spin from fixed quantum spin — Spin ½ and 1: Holstein, Ross, Vaydia, Cachazo, Guevara, Bautista, Febres Cordero, Kraus, Lin, Ruf, Zeng; Damgaard, Haddad, Helset, … spin 5/2: Chiodaroli, Johansson, Pichini
  - $\text{spin-}k \longrightarrow S^{2k}$
- Chiral higher-spin gravity/massive higher-spin gauge symmetry — Cangemi, Chiodaroli, Johansson, Ochirov, Pichini, Skvortsov
- Extract (Compton/Raman) amplitudes from Teukolsky equation — Bautista, Huang, J-W Kim; Bautista, Bonelli, Iossa, Tanzini, Zhou; Chen, Wang; Chen, Hsieh, Y-t Huang, J-W Kim
- Background field calculations — Kosmopoulos, Solon; Cheung, Parra-Martinez, Rothstein, Shah, Wilson-Gerow
- 4d EFT for arbitrary-spin particles in classical limit — Bern, Luna, RR, Shen, Zeng; + Teng; + Kosmopoulos, Scheopner, Vines; + Alaverdian

Spinning objects

Classical Hamiltonian (2PM):

$$H = E_1 + E_2 + \sum_a \sum_{n=1}^{\infty} \frac{G^2}{|\mathbf{r}|^2} c_n^a(\mathbf{p}^2)\Sigma_a \qquad \Sigma_a \in \left\{1, \frac{\mathbf{S}\cdot(\mathbf{r}\times\mathbf{p})}{|\mathbf{r}|^2}, \frac{(\mathbf{r}\cdot\mathbf{S})^2}{|\mathbf{r}|^4}, \frac{|\mathbf{S}|^2}{|\mathbf{r}|^2}, \dots\right\}$$

Field-theory discription: higher-spin quantum field theory ($\phi_s \equiv \phi_{a_1 a_2 \dots a_s}$)

$$\mathcal{L} = -\frac{1}{2}\phi_s(\nabla^2 + m^2)\phi_s \dots \qquad \nabla_\mu \phi_s = \partial_\mu \phi_s + \frac{i}{2}\omega_{\mu ab} M^{ab} \phi_s$$

Classical spin: expectation value of Lorentz generator $M^{\mu\nu}$ in the large-s spin coherent state (highest weight state along the direction of p)

$$\epsilon^s(p) \cdot M^{\mu\nu} \cdot \epsilon^s(p) = S^{\mu\nu} \qquad p_\mu S^{\mu\nu} = 0$$

Bern, Luna, Roiban, Shen, Zeng, 2005.03071
Bern, Kosmopoulos, Luna, Roiban, **FT**, 2203.06202
Bern, Kosmopoulos, Luna, Roiban, Scheopner, **FT**, Vines, 2308.14176
Alaverdian, Bern, Kosmopoulos, Luna, Roiban, Scheopner, **FT**, 2407.10928, 2503.03739

Spinning objects

It is straightforward to encode transitions to lower spin states $\qquad \mathbb{S}^a = (-i/2m)\epsilon^{abcd} M_{cd} \nabla_b$

$$\mathcal{L} = -\frac{1}{2}\phi_s(\nabla^2 + m^2)\phi_s + \frac{1}{8}R_{abcd}\phi_s M^{ab} M^{cd}\phi_s - \frac{C_2}{2m^2} R_{af_1bf_2}\nabla^a\phi_s \mathbb{S}^{(f_1}\mathbb{S}^{f_2)}\nabla^b\phi_s$$
$$+ \frac{D_2}{2m^2} R_{abcd}\nabla_i\phi_s\{M^{ai}M^{cd}\}\phi_s + \frac{E_2 - 2D_2}{2m^4} R_{abcd}\nabla^{(a}\nabla^{i)}\phi_s\{M^b{}_i M^d{}_j\}\nabla^{(c}\nabla^{j)}\phi_s + \mathcal{O}(M^3)$$

- The ghost-like nature of lower spin states can be cured through an analytic continuation at the classical level
- **Classical limit:** expectation value of Lorentz generator $M^{\mu\nu}$ in the large-s spin coherent state $\mathcal{E}_{\mu_1\ldots\mu_s} = \varepsilon^{(s)}_{\mu_1\ldots\mu_s} + u_{(\mu_1}\varepsilon^{(s-1)}_{\mu_2\ldots\mu_s)} + \ldots$, where $\varepsilon^{(s)}$ is the highest weight state along the direction of u, **[2308.14176]**

$$\mathcal{E}\cdot M^{ab}\cdot\mathcal{E} \to \mathsf{S}^{ab} \qquad\qquad \mathsf{S}^{ab} = S^{ab} + i(u^a K^b - u^b K^b)$$
$$\mathcal{E}\cdot\{M^{ab}M^{cd}\}\cdot\mathcal{E} \to \mathsf{S}^{ab}\mathsf{S}^{cd} \qquad\qquad S^{ab}u_b = K^a u_a = 0$$

[Similar coherent sum considered in Aoude, Ochirov, 2108.01649, etc]

- We identify K^a as an additional classical dynamical DOF
- When matching to a classical effective Hamiltonian, we identify $\mathbf{K} \equiv i\boldsymbol{K}$

Effective Hamiltonian through matching [2407.10928, 2503.03739]

Consider canonical spin in the COM frame

$$H = \sqrt{\mathbf{p}^2 + m_1^2} + \sqrt{\mathbf{p}^2 + m_2^2} + \sum_a \sum_{n=1}^{\infty} \left(\frac{G}{|\mathbf{r}|}\right)^n c_n^a(\mathbf{p}^2)\Sigma_a$$

Up to $\mathcal{O}(\mathsf{S}^2)$, the operators Σ_a takes value in

$$\begin{array}{ccc}
\mathbf{1} & \left((\mathbf{r}\times\mathbf{p})\cdot\mathbf{S}\right)/\mathbf{r}^2 & (\mathbf{r}\cdot\mathbf{K})/\mathbf{r}^2 \\
(\mathbf{r}\cdot\mathbf{S})^2/\mathbf{r}^4 & (\mathbf{r}\cdot\mathbf{K})\left((\mathbf{r}\times\mathbf{p})\cdot\mathbf{S}\right)/\mathbf{r}^4 & (\mathbf{r}\cdot\mathbf{K})^2/\mathbf{r}^4 \\
\mathbf{S}^2/\mathbf{r}^2 & \left(\mathbf{K}\cdot(\mathbf{p}\times\mathbf{S})\right)/\mathbf{r}^2 & \mathbf{K}^2/\mathbf{r}^2 \\
(\mathbf{p}\cdot\mathbf{S})^2/\mathbf{r}^2 & (\mathbf{r}\cdot\mathbf{S})\left((\mathbf{r}\times\mathbf{K})\cdot\mathbf{p}\right)/\mathbf{r}^4 & (\mathbf{p}\cdot\mathbf{K})^2/\mathbf{r}^2
\end{array}$$

We have computed the coefficients c_n^a up to $\mathcal{O}(\mathsf{S}^4)$ and $\mathcal{O}(G^2)$:

- c_0^a matches to the tree-level two-body amplitude
- c_1^a matches to the triangle coefficients of one-loop amplitude
- All the c_n^a coefficients are local functions of $\mathbf{p}^2$
- The coefficient of $(\mathbf{r}\cdot\mathbf{K})/\mathbf{r}^2$ vanishes [the $\mathbf{K}$ DOF only induces quadrupolar interactions]

Generic spinning bodies [2407.10928, 2503.03739]

$$\mathcal{L} = -\frac{1}{2}\phi_s(\nabla^2 + m^2)\phi_s + \frac{1}{8}R_{abcd}\phi_s M^{ab}M^{cd}\phi_s - \frac{C_2}{2m^2}R_{af_1bf_2}\nabla^a\phi_s\mathbb{S}^{(f_1}\mathbb{S}^{f_2)}\nabla^b\phi_s$$
$$+ \frac{D_2}{2m^2}R_{abcd}\nabla_i\phi_s\{M^{ai}M^{cd}\}\phi_s + \frac{E_2 - 2D_2}{2m^4}R_{abcd}\nabla^{(a}\nabla^{i)}\phi_s\{M^b{}_iM^d{}_j\}\nabla^{(c}\nabla^{j)}\phi_s$$

- The C_2-operator is the only independent operator assuming that rest frame spin is the only dynamical degree of freedom Porto, gr-qc/0511061; Levi, Steinhoff, 1501.04956
- The D_2- and E_2-operators supply additional $\mathcal{O}(SK)$ and $\mathcal{O}(K^2)$ interactions

$D_2 = E_2 = 0 \implies$ Conventional compact object described by $H(\mathbf{r},\mathbf{p},\mathbf{S})$

$C_2 = D_2 = E_2 = 0 \implies$ Kerr black hole [Cangemi's talk]

Generic values: compact object described by the conservative $H(\mathbf{r},\mathbf{p},\mathbf{S},\mathbf{K})$

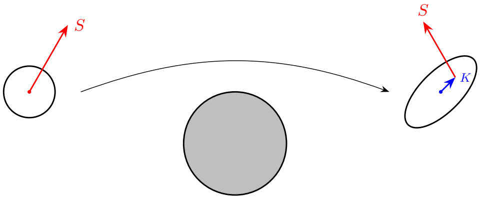
Example system with the extra DOF

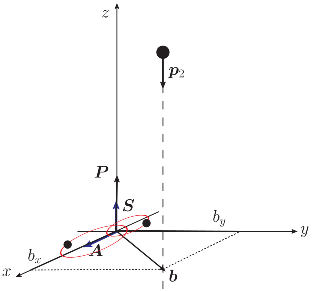
Scattering of a Newtonian bound state

$$H = \frac{\boldsymbol{p}_2^2}{2m_2} + \frac{\boldsymbol{P}^2}{2m_1} + H_0(\boldsymbol{p},\boldsymbol{r}) - \frac{Gm_{B1}m_2}{|\boldsymbol{R} + \frac{m_{B2}}{m_1}\boldsymbol{r}|} - \frac{Gm_{B2}m_2}{|\boldsymbol{R} - \frac{m_{B1}}{m_1}\boldsymbol{r}|}$$

$$= \frac{\boldsymbol{p}_2^2}{2m_2} + \frac{\boldsymbol{P}^2}{2m_1} + H_0(\boldsymbol{p},\boldsymbol{r}) - \frac{Gm_1m_2}{|\boldsymbol{R}|} - \frac{3G\mu_B m_2}{2|\boldsymbol{R}|^5}\underbrace{\left((\boldsymbol{r}\cdot\boldsymbol{R})^2 - \frac{1}{3}|\boldsymbol{r}|^2|\boldsymbol{R}|^2\right)}_{Q_{ij}(\boldsymbol{r})Q^{ij}(\boldsymbol{R})}$$

where $m_1 = m_{B1} + m_{B2}$ and $\mu_B = m_{B1}m_{B2}/m_1$





Example system with the extra DOF

Scattering of a Newtonian bound state

Transition amplitude (in and out state are both elliptical orbit coherent states)

$$\mathcal{A}_{\mathrm{i}\to\mathrm{f}} = \frac{3G\mu_B m_2 r_{\mathrm{cl},n}^2}{2|\boldsymbol{b}|^2 v_0}\left[\frac{2(\boldsymbol{b}\cdot\boldsymbol{A})^2}{|\boldsymbol{b}|^2} - |\boldsymbol{A}|^2\right] \iff \mathcal{M}^{2\text{ body}} = A_1\mathbf{K}^2 + A_2(\boldsymbol{b}\cdot\mathbf{K})^2$$

Amplitude matching gives:

- Spin $\Leftrightarrow$ bound system total orbital angular momentum
- $\mathbf{K}$-vector $\Leftrightarrow$ Laplace-Runge-Lenz vector

$$\mathbf{K} = Gm_1^2\frac{\mu_B}{m_1}\sqrt{\frac{\mu_B}{2|\mathsf{E}_{\mathrm{i}}^B|}}\,\boldsymbol{A}$$

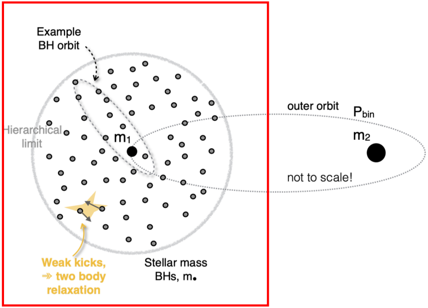
- Wilson coefficient

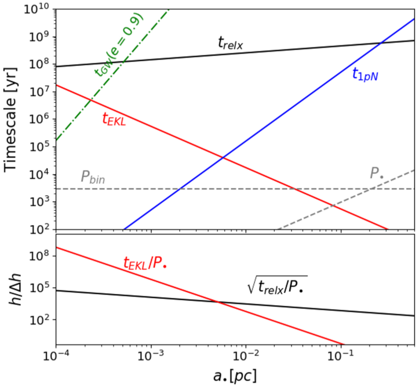
$$E_2^{\text{bound 2-body}} = \frac{3m_1}{4|\mathsf{E}_{\mathrm{i}}^B|}$$

field theory Lagrangian $\mathcal{L} \sim \frac{E_2}{2m^4}R_{abcd}\nabla^{(a}\nabla^{i)}\phi_s\{M^b{}_i M^d{}_j\}\nabla^{(c}\nabla^{j)}\phi_s$

Possible application: elliptical Kozai-Lidov

Work in progress + Smadar Naoz





EKL: eccentricity perturbation due to the third SMBH

$t_{\rm 1pN}$:	precession time scale	$t_{\rm EKL}$:	EKL time scale
$t_{\rm GW}$:	circularization due to GW rad.	$t_{\rm relx}$:	scattering relaxation
$P_{\rm bin}$:	SMBH binary period	$P_\bullet$:	period of $m_\bullet$ around m_1

Naoz, etc, 2202.12303

Table of Contents

KMOC formalism Kosower, Maybee, O'Connell, 1811.10950

Observable in a scattering process as an in-in correlator

$$\begin{aligned}\Delta O &= \langle\psi_{\text{out}}|\mathbb{O}|\psi_{\text{out}}\rangle - \langle\psi_{\text{in}}|\mathbb{O}|\psi_{\text{in}}\rangle \\ &= \langle\psi_{\text{in}}|\hat{S}^\dagger\mathbb{O}\hat{S}|\psi_{\text{in}}\rangle - \langle\psi_{\text{in}}|\mathbb{O}|\psi_{\text{in}}\rangle \\ &= i\langle\psi_{\text{in}}|[\mathbb{O},\hat{T}]|\psi_{\text{in}}\rangle + \langle\psi_{\text{in}}|\hat{T}^\dagger[\mathbb{O},\hat{T}]|\psi_{\text{in}}\rangle\end{aligned}$$

$$\hat{S} = 1 + i\hat{T}$$
$$\hat{T} - \hat{T}^\dagger = i\hat{T}^\dagger\hat{T}$$

Incoming state: $|\psi_{\text{in}}\rangle = \int \mathrm{d}\Phi[p_1]\mathrm{d}\Phi[p_2]\phi(p_1)\phi(p_2)e^{ip_1\cdot b_1 + ip_2\cdot b_2}|p_1 p_2\rangle$

Single particle phase-space and wavefunction:

$$\mathrm{d}\Phi[p] = \frac{\mathrm{d}^4 p}{(2\pi)^4}2\pi\Theta(p^0)\delta(p^2 - m^2) \qquad \int \mathrm{d}\Phi[p]|\phi(p)|^2 = 1$$

We can compute observables by dressing amplitudes and cuts with the corresponding operators

$$\Delta O = \int \mathrm{d}\Phi[p_1]\mathrm{d}\Phi[p_2]\mathrm{d}\Phi[p_1']\mathrm{d}\Phi[p_2']\phi(p_1)\phi(p_2)\phi(p_1')\phi(p_2')e^{i(p_1-p_1')\cdot b_1 + i(p_2-p_2')\cdot b_2}$$
$$\times\Big[i\langle p_1'p_2'|[\mathbb{O},\hat{T}]|p_1p_2\rangle + \langle p_1'p_2'|\hat{T}^\dagger[\mathbb{O},\hat{T}]|p_1p_2\rangle\Big]$$

We can choose $\mathbb{O}$ to be $\underbrace{P^\mu}_{\text{impulse}}$, $\underbrace{K^\mu}_{\text{energy loss}}$, $\underbrace{h_{\mu\nu}}_{\text{waveform}}$, $\underbrace{\mathcal{E}(n)}_{\text{ANEC}}$, etc

Classical observables Kosower, Maybee, O'Connell, 1811.10950

The wave packets are highly localized, while q_i is much smaller than their spread

$$\int \mathrm{d}\Phi[p_1]\mathrm{d}\Phi[p_2]\mathrm{d}\Phi[p_1']\mathrm{d}\Phi[p_2']\phi(p_1)\phi(p_2)\phi^*(p_1')\phi^*(p_2')e^{i(p_1-p_1')\cdot b_1+i(p_2-p_2')\cdot b_2}$$

$$\simeq \int \frac{\mathrm{d}^d q_1}{(2\pi)^d}\frac{\mathrm{d}^d q_2}{(2\pi)^d}\hat{\delta}(2\bar{p}_1\cdot q_1)\hat{\delta}(2\bar{p}_2\cdot q_2)e^{iq_1\cdot b_1+iq_2\cdot b_2} \qquad \hat{\delta}(x)=2\pi\delta(x)$$

Resolution of identity in the two-massive-particle subspace

$$\mathbb{I}=\sum_X\int \mathrm{d}\Phi[r_1]\mathrm{d}\Phi[r_2]|r_1r_2X\rangle\langle r_1r_2X|$$

Expansion in the soft region $(q_i,\ell,k,G)\to(\hbar q_i,\hbar\ell,\hbar k,\hbar^{-1}G)$

Classical observables

$$\Delta O_{\mathrm{cl}}=\int \frac{\mathrm{d}^d q_1}{(2\pi)^d}\frac{\mathrm{d}^d q_2}{(2\pi)^d}\hat{\delta}(2\bar{p}_1\cdot q_1)\hat{\delta}(2\bar{p}_2\cdot q_2)e^{iq_1\cdot b_1+iq_2\cdot b_2}\left[i\langle p_1'p_2'|[\mathbb{O},\hat{T}]|p_1p_2\rangle+\langle p_1'p_2'|\hat{T}^\dagger[\mathbb{O},\hat{T}]|p_1p_2\rangle\right]_{\mathrm{cl}}$$

Waveform and metric perturbation

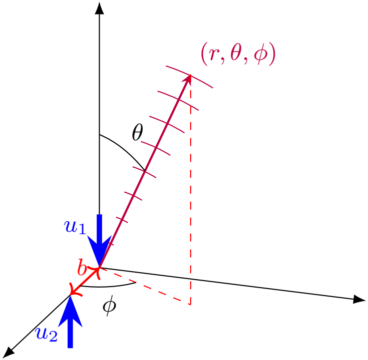
Herderschee, Roiban, **FT**, 2303.06112

Bini, Damour, De Angelis, Geralico, Herderschee, Roiban, **FT**, 2402.06604

De Angelis, Herderschee, Roiban, **FT**, 2511.10637

Metric perturbation: $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$

Waveform: $W(T_{\mathrm{R}}, \theta, \phi) = \dfrac{1}{4G} \lim_{r\to\infty} r\varepsilon_+^\mu \varepsilon_+^\nu h_{\mu\nu} = \dfrac{1}{4G}(h_+^\infty - i h_\times^\infty)$



Waveform

Cristofoli, Gonzo, Kosower, O'Connell, 2107.10193

Operator: metric perturbation

$$\mathbb{H} = \varepsilon_+^\mu \varepsilon_+^\nu \hat{h}_{\mu\nu}(x) = \int \mathrm{d}\Phi[k] \left[\hat{a}_{--}(k)\, e^{-ik\cdot x} + \mathrm{c.c} \right]$$

Since there is no radiation at $t \to -\infty$,

$$\Delta H(x) = \langle \psi_{\mathrm{in}} | \hat{S}^\dagger \mathbb{H} \hat{S} | \psi_{\mathrm{in}} \rangle = \int \mathrm{d}\Phi[k] \left[\tilde{J}(k) e^{-ik\cdot x} - \mathrm{c.c} \right] \qquad \tilde{J}(k) = \langle \psi_{\mathrm{in}} | \hat{S}^\dagger \hat{a}_{--}(k) \hat{S} | \psi_{\mathrm{in}} \rangle$$

Assuming the spatial current $J(y)$ is localized around $y = 0$,

$$\Delta H(x) \Big|_{|\boldsymbol{x}| \to \infty} = \frac{1}{4\pi |\boldsymbol{x}|} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} W(\omega, \boldsymbol{n}) e^{-i\omega\tau} \qquad (\tau = x^0 - |\boldsymbol{x}| : \text{ retarded time})$$

$$\mathcal{W}(\varepsilon, \omega, \boldsymbol{n}, p_1, p_2, b_1, b_2) = (-i) \langle \psi_{\mathrm{in}} | \hat{S}^\dagger \hat{a}_{--}(k) \hat{S} | \psi_{\mathrm{in}} \rangle \Big|_{k = (\omega, \omega \boldsymbol{n})}^{\text{IR finite}} \qquad \text{(frequency domain waveform)}$$

(The IR divergence can be absorbed in the definition of τ)

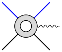
Relevant matrix elements for waveform

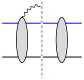
In the space of momentum transfer:

$$(-i)\langle\psi_{\rm in}|\hat{S}^\dagger \hat{a}(k)\hat{S}|\psi_{\rm in}\rangle = \int \frac{\mathrm{d}^d q_1}{(2\pi)^d}\frac{\mathrm{d}^d q_2}{(2\pi)^d}\hat{\delta}(2p_1\cdot q_1)\hat{\delta}(2p_2\cdot q_2)e^{iq_1\cdot b_1+iq_2\cdot b_2}$$
$$\times\left[\langle p_1' p_2' k|\hat{T}|p_1 p_2\rangle - i\langle p_1' p_2'|\hat{T}^\dagger \hat{a}(k)\hat{T}|p_1 p_2\rangle\right]$$

The matrix elements can be classified as follows:

$$\langle p_1' p_2' k|\hat{T}|p_1 p_2\rangle\Big|_{\rm conn} = \mathcal{M}^{\rm tree} + \mathcal{M}^{1\ \rm loop} \qquad i\langle p_1' p_2'|\hat{T}^\dagger \hat{a}(k)\hat{T}|p_1 p_2\rangle\Big|_{\rm conn} = \mathcal{S}^{1\ \rm loop}$$



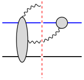


The full connected contribution to waveform:

$$\mathscr{M}_{\rm conn} = \mathcal{M}^{\rm tree} + \mathcal{M}^{1\ \rm loop} - \mathcal{S}^{1\ \rm loop}$$

Contribution from disconnected T matrix elements:

$$\langle p_1' p_2' k|\hat{T}|p_1 p_2\rangle\Big|_{\rm disc} = \mathscr{M}_{\rm const} \qquad i\langle p_1' p_2'|\hat{T}^\dagger \hat{a}(k)\hat{T}|p_1 p_2\rangle\Big|_{\rm disc} = \mathscr{M}_{\rm disc}$$



Tree-level amplitudes and LO waveform

From GR: Kovacs, Thorne, Astrophys. J. 224 62, 1978
From world line QFT: Jakobsen, Mogull, Plefka, Steinhoff, 2101.12688
From KMOC: Cristofoli, Gonzo, Kosower, O'Connell, 2107.10193

For convenience, we define the Fourier transform from q-space to b-space,

$$\mathrm{FT}\Big[f(q_1,q_2,k)\Big] = \int \frac{\mathrm{d}^d q_1}{(2\pi)^d}\frac{\mathrm{d}^d q_2}{(2\pi)^d}\hat{\delta}(2p_1\cdot q_1)\hat{\delta}(2p_2\cdot q_2)e^{iq_1\cdot b_1+iq_2\cdot b_2}\hat{\delta}(q_1+q_2-k)f(q_1,q_2,k)\Big|_{k=(\omega,\omega\boldsymbol{n})}$$

LO waveform from KMOC

$$\mathcal{W}(\omega,\boldsymbol{n})\Big|_{\mathrm{LO}} = \mathrm{FT}\Big[\mathcal{M}^{\mathrm{tree}}\Big]$$

Classical tree-level amplitude Luna, Nicholson, O'Connell, White, 1711.03901

$$\begin{aligned}\mathcal{M}^{\mathrm{tree}} &= -\kappa^2 m_1^2 m_2^2\left[\frac{(P_{12}\cdot\varepsilon)^2+\sigma(Q_{12}\cdot\varepsilon)(P_{12}\cdot\varepsilon)}{q_1^2 q_2^2}+\left(\sigma^2-\frac{1}{d-2}\right)\left(\frac{(Q_{12}\cdot\varepsilon)^2}{4q_1^2q_2^2}-\frac{(P_{12}\cdot\varepsilon)^2}{4w_1^2w_2^2}\right)\right]\\ &= \mathcal{M}^{\mathrm{tree}}_{d=4}+\epsilon\mathcal{M}^{\mathrm{tree}}_{\mathrm{extra}}\end{aligned}$$

$$P_{12}^\mu = w_1 u_2^\mu - w_2 u_1^\mu \qquad u_i = p_i/m_i \qquad w_i = u_i\cdot k$$

$$Q_{12}^\mu = (q_1-q_2)^\mu - \frac{q_1^2}{w_1}u_1^\mu + \frac{q_2^2}{w_2}u_2^\mu$$

Of course, at LO the $\mathcal{O}(\epsilon)$ term is irrelevant.

Compact analytic result

Time domain waveform [following the conventions in De Angelis, Novichikov, Gonzo, 2309.17429]

$$W^{(1)}(\tau,\theta,\phi) = -\frac{Gm_1m_2}{4\sqrt{-b^2}\,\hat{w}_1^2\hat{w}_2^2\sqrt{1+T_2^2}\left[\sigma+\sqrt{(1+T_1^2)(1+T_2^2)}+T_1T_2\right]}$$

$$\times\left\{\frac{3\hat{w}_1+2\sigma(2T_1T_2\hat{w}_1-T_2^2\hat{w}_2+\hat{w}_2)-(2\sigma^2-1)\hat{w}_1}{\sigma^2-1}\mathcal{F}_{1,2}^2\right.$$

$$-\frac{4\sigma T_2\hat{w}_2\mathcal{F}_1+2(2\sigma^2-1)\left[T_1(1+T_2^2)\hat{w}_2\mathcal{F}_1+T_2(T_1T_2\hat{w}_1+\hat{w}_2)\mathcal{F}_2\right]}{\sqrt{\sigma^2-1}}\mathcal{F}_{1,2}$$

$$\left.+4(1+T_2^2)\hat{w}_2\mathcal{F}_1\mathcal{F}_2-4\sigma(1+T_2)^2\hat{w}_2(\mathcal{F}_1^2+\mathcal{F}_2^2)+2(2\sigma^2-1)(1+2T_2^2)\hat{w}_2\mathcal{F}_1\mathcal{F}_2\right\}+(1\leftrightarrow 2)$$

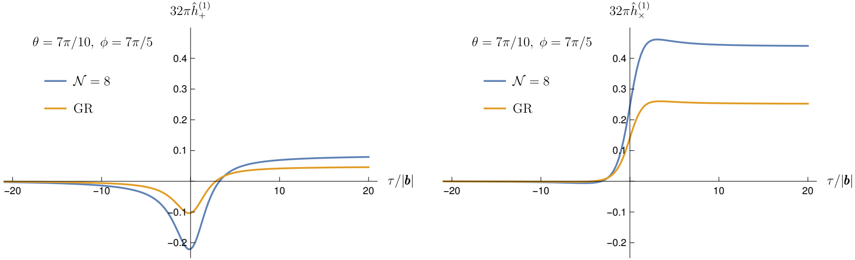
where

$$\hat{w}_1 = u_1\cdot n \qquad \hat{w}_2 = u_2\cdot n \qquad T_1 = \frac{\sqrt{\sigma^2-1}\,(\tau-b_1\cdot n)}{\hat{w}_1\sqrt{-b^2}} \qquad T_2 = \frac{\sqrt{\sigma^2-1}\,(\tau-b_2\cdot n)}{\hat{w}_2\sqrt{-b^2}}$$

$$b^\mu = b_1^\mu - b_2^\mu \qquad \hat{b}^\mu = \frac{b^\mu}{\sqrt{-b^2}}$$

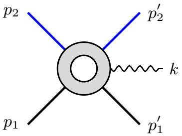
$$\mathcal{F}_{1,2} = (\varepsilon_+^*\cdot u_1)\hat{w}_2 - (\varepsilon_+^*\cdot u_2)\hat{w}_1 \qquad \mathcal{F}_1 = (\hat{b}\cdot\varepsilon_+^*)\hat{w}_1 - (\hat{b}\cdot n)(\varepsilon_+^*\cdot u_1) \qquad \mathcal{F}_2 = (\hat{b}\cdot\varepsilon_+^*)\hat{w}_2 - (\hat{b}\cdot n)(\varepsilon_+^*\cdot u_2)$$

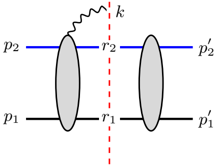
LO waveform



Agree with Jakobsen, Mogull, Plefka, Steinhoff, 2101.12688
From spinning binaries: De Angelis, Novichkov, Gonzo, 2309.17429
Brandhuber, Brown, Chen, Gowdy, Travaglini, 2310.04405
Aoude, Haddad, Heissenberg, Helset, 2310.05832

One-loop matrix elements Herderschee, Roiban, **FT**, 2303.06112





- Use generalized unitarity to construct the integrand
- Expand the integrand in the classical limit

$$\frac{1}{(p+\ell)^2 - m^2 + i0} = \frac{1}{2p\cdot\ell + \ell^2 + i0} = \frac{1}{2p\cdot\ell + i0}\sum_{n=0}^{\infty}\left(-\frac{\ell^2}{2p\cdot\ell + i0}\right)^n$$

- Reduce the integrand to master integral basis [IBP automated by FIRE6: Smirnov, Chuharev, 1901.07808]
- Classically singular (superclassical) terms cancel between the two matrix elements
- The cut part makes all matter propagators retarded

$$\text{amplitude:}\quad \mathrm{PV}\frac{1}{2u_1\cdot\ell} \xrightarrow{\text{add cut}} \frac{1}{2u_1\cdot\ell - i0}$$

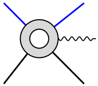
- Classical waveform contains only imaginary IR divergences

Equivalence to world-line formalism proved by Capatti, Zeng, 2412.10864

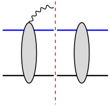
The full connected contribution

Herderschee, Roiban, **FT**, 2303.06112
Brandhuber, Brown, Chen, De Angelis, Gowdy, Travaglini, 2303.06111
Georgoudis, Heissenberg, Vazquez-Holm, 2303.07006

$$\begin{aligned}
\mathcal{M}^{1\text{ loop}} =& -\frac{i\,\kappa^2}{32\pi}(m_1 w_1 + m_2 w_2)\left[\frac{1}{\epsilon} - \log\frac{w_1 w_2}{\mu^2}\right]\mathcal{M}^{\text{tree}}_{d=4} \\
&+ \kappa^4\left[A^R_{\text{rat}} + \frac{A^R_1}{\sqrt{w_2^2 - q_1^2}} + \frac{A^R_2}{\sqrt{w_1^2 - q_2^2}} + \frac{A^R_3}{\sqrt{-q_1^2}} + \frac{A^R_4}{\sqrt{-q_2^2}}\right] \\
&+ i\,\kappa^4\left[A^I_{\text{rat}} + A^I_1\frac{\operatorname{arcsinh}\frac{w_2}{\sqrt{-q_1^2}}}{\sqrt{w_2^2 - q_1^2}} + A^I_2\frac{\operatorname{arcsinh}\frac{w_1}{\sqrt{-q_2^2}}}{\sqrt{w_1^2 - q_2^2}} + A^I_3\log\frac{q_2^2}{q_1^2} + A^I_4\log\frac{w_1}{w_2} + A^I_5\frac{\operatorname{arccosh}\sigma}{(\sigma^2-1)^{3/2}}\right] \\
=& -\frac{i\,\kappa^2}{32\pi\epsilon}(m_1 w_1 + m_2 w_2)\mathcal{M}^{\text{tree}}_{d=4} + \widetilde{\mathcal{M}}^{\text{fin}}_{1\text{ loop}}
\end{aligned}$$



$$\begin{aligned}
\mathcal{S}^{1\text{ loop}} =& \frac{i\,\kappa^2\Gamma}{64\pi}(m_1 w_1 + m_2 w_2)\left[\frac{1}{\epsilon} - \log\frac{w_1 w_2}{(\sigma^2-1)\mu^2}\right]\mathcal{M}^{\text{tree}}_{d=4} \qquad \Gamma = \frac{3\sigma - 2\sigma^3}{(\sigma^2-1)^{3/2}} \\
&+ \frac{i\,\kappa^4}{256\pi}\frac{(2\sigma^2-1)^2}{(\sigma^2-1)^{3/2}}(m_1 w_1 + m_2 w_2)\left[\frac{1}{\epsilon} - \log\frac{w_1 w_2}{(\sigma^2-1)\mu^2}\right]\frac{m_1^2 m_2^2 (u_1\cdot f\cdot u_2)^2(w_1^2 + \sigma w_1 w_2 + w_2^2)}{w_1^3 w_2^3} \\
&+ i\,\kappa^4\left[A^{\text{cut}}_{\text{rat}} + A^{\text{cut}}_1\log\frac{w_1}{w_2} + A^{\text{cut}}_2\log\frac{w_1 w_2}{-q_1^2} + A^{\text{cut}}_3\log\frac{w_1 w_2}{-q_2^2} + A^{\text{cut}}_4\frac{\operatorname{arccosh}\sigma}{(\sigma^2-1)^{3/2}}\right] \\
=& \frac{i\,\kappa^2\Gamma}{64\pi\epsilon}(m_1 w_1 + m_2 w_2)\mathcal{M}^{\text{tree}}_{d=4} + \widetilde{\mathcal{S}}^{\text{fin}}_{1\text{ loop}} + \mathcal{S}^{\text{local}}_{1\text{ loop}}
\end{aligned}$$



The full connected contribution

The IR divergence exponentiates Caron-Huot, Giroux, Hannesdottir, Mizera, 2308.02125

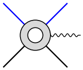
$$\mathscr{M}_{\text{conn}} = \mathscr{M}_{\text{conn}}^{\text{fin}} \exp\left[-i\omega \frac{\kappa^2 E(1-\Gamma/2)}{32\pi\epsilon}\right] \qquad (m_1 w_1 + m_2 w_2 = \omega E)$$

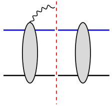
such that it can be absorbed by a re-definition of retarded time

$$\tau \to \tau - \frac{\kappa^2 E(1-\Gamma/2)}{32\pi\epsilon}$$

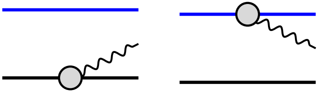
At one-loop level, this leads to

$$\begin{aligned}
\mathscr{M}_{\text{conn}}^{\text{fin}} &= \mathcal{M}^{\text{tree}} + \mathcal{M}^{1\text{ loop}} - \mathcal{S}^{1\text{ loop}} + \frac{i\kappa^2(m_1 w_1 + m_2 w_2)(1-\Gamma/2)}{32\pi\epsilon}\mathcal{M}^{\text{tree}} \\
&= \mathcal{M}^{\text{tree}} + \left[\widetilde{\mathcal{M}}_{1\text{ loop}}^{\text{fin}} + \frac{i\kappa^2(m_1 w_1 + m_2 w_2)\mathcal{M}_{\text{extra}}^{\text{tree}}}{32\pi}\right] - \left[\widetilde{\mathcal{S}}_{1\text{ loop}}^{\text{fin}} + \frac{i\kappa^2\Gamma(m_1 w_1 + m_2 w_2)\mathcal{M}_{\text{extra}}^{\text{tree}}}{64\pi}\right] \\
&= \mathcal{M}^{\text{tree}} + \mathcal{M}_{1\text{ loop}}^{\text{fin}} - \mathcal{S}_{1\text{ loop}}^{\text{fin}}
\end{aligned}$$





Disconnected amplitude contribution



They are supported only by zero-energy gravitons

$$\mathcal{W}_{\rm const}(\omega,\boldsymbol{n}) = \int \mathrm{d}\Phi_{p_1}\mathcal{M}_3(p_1,k)e^{ik\cdot b_1} + \int \mathrm{d}\Phi_{p_2}\mathcal{M}_3(p_2,k)e^{ik\cdot b_2} = \hat{\delta}(\omega)\left[\frac{m_1(\varepsilon\cdot u_1)^2}{u_1\cdot n} + \frac{m_2(\varepsilon\cdot u_2)^2}{u_2\cdot n}\right]$$

The constant background agrees with the MPM calculation

$$\mathcal{W}_{\rm const}(\tau,\boldsymbol{n}) = \frac{m_1(\varepsilon\cdot u_1)^2}{u_1\cdot n} + \frac{m_2(\varepsilon\cdot u_2)^2}{u_2\cdot n} \qquad n=(1,\boldsymbol{n})$$

It can be removed by a BMS supertranslation Veneziano, Vilkovisky, 2201.11607

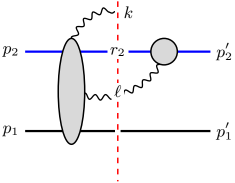
$$\delta_T\left[\frac{h^\infty_{ab}}{r}\right] = \frac{1}{r}(2D_aD_b - \gamma_{ab}\Delta)T(n) - T(n)\partial_\tau\left[\frac{h^\infty_{ab}}{r}\right] \qquad T(n) = 2G\left[m_1\frac{w_1}{\omega}\log\frac{w_1}{\omega} + m_2\frac{w_2}{\omega}\log\frac{w_2}{\omega}\right]$$

The MPM waveform is in the intrinsic BMS frame. If we assume $\mathcal{M}_3 = 0$, then we are in the canonical BMS frame. They are related by the above BMS supertranslation.

Disconnected cut contribution [DB, TD, SDA, AG, AH, RR, **FT**, 2402.06604]

The T matrices in the KMOC cut can be disconnected:

$$i \langle p_1' p_2' | \hat{T}^\dagger \hat{a}(k) \hat{T} | p_1 p_2 \rangle \Big|_{\text{disc.}} \supset \int \mathrm{d}\Phi[\ell] \mathrm{d}\Phi[r_2]$$



- The cut graviton ℓ only has support on zero energy
- We only need to consider the soft limit of the left blob

$$\mathcal{M}_6 \simeq \kappa \, \mathcal{M}_5 \left[\sum_a \frac{\eta_a \varepsilon_{\mu\nu}(\ell) p_a^\mu p_a^\nu}{2\ell \cdot p_a + i0\eta_a} + \frac{\varepsilon_{\mu\nu}(\ell) k^\mu k^\nu}{2\ell \cdot k + i0} \right]$$

- This loop integral requires a further regularization to include the zero-energy graviton contribution

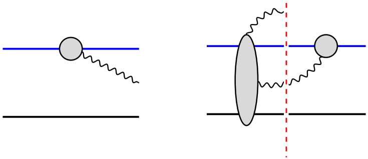
Disconnected cut contribution [DB, TD, SDA, AG, AH, RR, **FT**, 2402.06604]

The disconnected T matrix elements in the cut lead to

$$\mathscr{M}_{\rm disc} = i\langle p_1' p_2' | \hat{T}^\dagger \hat{a}(k) \hat{T} | p_1 p_2 \rangle \Big|_{\rm disc} = -2i\omega G \left[m_1 \frac{w_1}{\omega} \log \frac{w_1}{\omega} + m_2 \frac{w_2}{\omega} \log \frac{w_2}{\omega} \right] \mathcal{M}^{\rm tree} = -i\omega T(n) \mathcal{M}^{\rm tree}$$

which exactly reproduces the iteration part of the BMS supertranslation

$$\delta_T \left[\frac{h_{ab}^\infty}{r} \right] = \frac{1}{r}(2D_a D_b - \gamma_{ab}\Delta) T(n) - T(n)\partial_\tau \left[\frac{h_{ab}^\infty}{r} \right]$$



This contribution can be re-summed into a coherent zero-energy graviton dressing of asymptotic massive states Elkhidir, O'Connell, Roiban, 2408.15961 and WIP

[O'Connell's talk]

Full KMOC waveform at one-loop [DB, TD, SDA, AG, AH, RR, **FT**, 2402.06604]

$$\mathcal{W}^{\mathrm{KMOC}}(\omega, \boldsymbol{n}) = \mathcal{W}_{\mathrm{const}} + \mathcal{W}^{\mathrm{tree}} + \mathrm{FT}\Big[\mathcal{M}_{1\text{ loop}}^{\mathrm{fin}} - \mathcal{S}_{1\text{ loop}}^{\mathrm{fin}} + \mathscr{M}_{\mathrm{disc}}\Big]$$

- Fourier transform to the impact-parameter space can only be done numerically for generic kinematic setup (see Brunello, De Angelis 2403.08009 for improvements)
- Analytic computation is available under the soft graviton limit and/or small relative velocity (PN) expansion

Soft expansion (up to the first non-universal coefficient)

$$\mathcal{W}^{\mathrm{KMOC}}(\omega, \boldsymbol{n}) \sim \frac{\mathcal{A}}{\omega} + \mathcal{B}\log\omega + \mathcal{C}\omega(\log\omega)^2 + \mathcal{D}\omega\log\omega$$

All order conjecture in leading log: Alessio, Di Vecchia, Heissenberg, 2407.04128

PN expansion (up to 2.5PN)

$M = m_1 + m_2$

$\nu = m_1 m_2 / M^2$

$p_\infty = \sqrt{\sigma^2 - 1}$

$$\mathcal{W}^{\mathrm{KMOC}}(\omega, \boldsymbol{n}) \sim 1 + \frac{GM^2\nu}{p_\infty}(1 + p_\infty + p_\infty^2 + p_\infty^3 + p_\infty^4 + p_\infty^5 + \ldots)$$
$$+ \frac{GM}{bp_\infty^2}\frac{GM^2\nu}{p_\infty}(1 + p_\infty + p_\infty^2 + p_\infty^3 + p_\infty^4 + p_\infty^5 + \ldots)$$

Full KMOC waveform at one-loop [DB, TD, SDA, AG, AH, RR, **FT**, 2402.06604]

$$\mathcal{W}^{\mathrm{KMOC}}(\omega,\boldsymbol{n}) = \mathcal{W}_{\mathrm{const}} + \mathcal{W}^{\mathrm{tree}} + \mathrm{FT}\Big[\mathcal{M}^{\mathrm{fin}}_{1\ \mathrm{loop}} - \mathcal{S}^{\mathrm{fin}}_{1\ \mathrm{loop}} + \mathscr{M}_{\mathrm{disc}}\Big]$$

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Both in perfect agreement with
the GR based multipolar post-Minkowskian calculation

Bini, Damour, Geralico, 2309.14925

Similar comparison up to $\mathcal{O}(p_\infty^3)$ is done in Georgoudis, Heissenberg, Russo, 2402.06361

Soft expansion (up to the first non-universal coefficient)

$$\mathcal{W}^{\mathrm{KMOC}}(\omega,\boldsymbol{n}) \sim \frac{\mathcal{A}}{\omega} + \mathcal{B}\log\omega + \mathcal{C}\omega(\log\omega)^2 + \mathcal{D}\omega\log\omega$$

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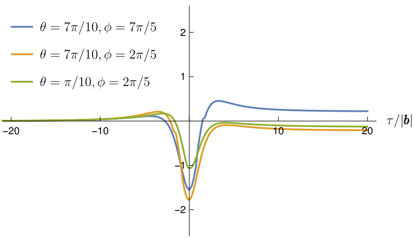
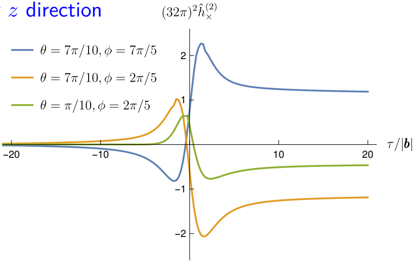
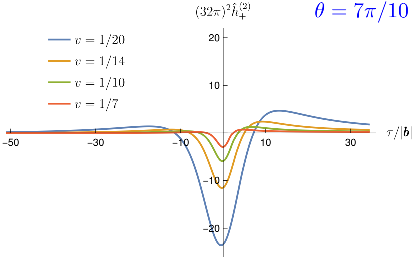
NLO waveform

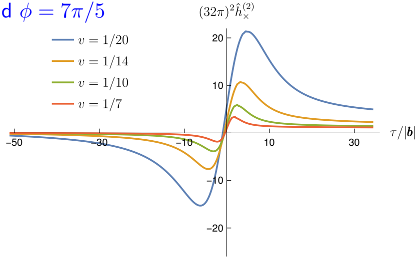
$v = 1/5$ along z direction





$\theta = 7\pi/10$ and $\phi = 7\pi/5$





Current status

Comparison between MPM and EFT:

- Recently, the agreement between MPM and EFT result has been pushed to 3.5PN (or p_∞^7) Bini, Damour, Geralico, 2604.21522 [Damour's talk]
- The agreement requires an additional BMS supertranslation on top of the Veneziano-Vilkovisky one

Further development:

- Improved IBP reduction: treat loop integral and Fourier transform on the same footing Brunello, De Angelis, Kosower, 2511.05412
- New integration technique using intersection theory Brunello, Chestnov, Crisanti, Giroux, Smith, 2510.26874
- Quadratic-in-spin waveform Bohnenblust, Ita, Kraus, Schlenk, 2505.15724, 2312.14859
- 2PN at two-loop order Bini, Damour, Geralico, 2407.02076
- Universal soft coefficient to all order Alessio, Di Vecchia, Heissenberg, 2407.04128
- and more ......

Off-shell extension: Newman-Penrose scalar

De Angelis, Herderschee, Roiban, **FT**, 2511.10637

Peeling theorem (requiring C^4 regularity at spatial infinity): $\Psi_n \sim r^{n-5}$ [Penrose 1965]

$$\Psi_4 = \Psi_{abcd}\lambda^a\lambda^b\lambda^c\lambda^d \qquad \Psi_3 = \Psi_{abcd}\lambda^a\lambda^b\lambda^c\rho^d \qquad \Psi_2 = \Psi_{abcd}\lambda^a\lambda^b\rho^c\rho^d$$
$$\Psi_1 = \Psi_{abcd}\lambda^a\rho^b\rho^c\rho^d \qquad \Psi_0 = \Psi_{abcd}\rho^a\rho^b\rho^c\rho^d$$

Metric perturbation at finite space-time distance

$$\begin{aligned} h_{\mu\nu}(x) &= \langle\Psi, t|\hat{h}_{\mu\nu}(x)|\Psi, t\rangle - \langle\Psi, -\infty|\hat{h}_{\mu\nu}(x)|\Psi, -\infty\rangle \\ &= -\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{e^{-ikx}}{(k^0+i0)^2-\boldsymbol{k}^2}T^{\mu\nu}_{\alpha\beta}J^{\alpha\beta}(k) \end{aligned}$$

In the soft graviton limit $k^\mu \to 0$, there are two relevant regions

- Radiation region: $k^\mu \sim r^0$
- Coulomb region: $k^\mu \sim r^{-1}$ [more scales, more challenging]

The Coulomb region gives normal peeling behavior (including the known violations), while the radiation region introduces novel peeling violations:

$$\Psi_0 \sim r^{-4} \qquad \Psi_1 \sim r^{-3} \quad \text{at } G^3 \text{ order}$$

[in addition to Damour 1985 and Christodoulou 2002] [still need further investigation; see Boschetti, Campiglia, 2603.22681]

Table of Contents

Recent developments: Amplitudes + X

- Re-summation:
  - Perturbative orders: eikonal, Magnusian, etc Di Vecchia, Heissenberg, Russo, Veneziano, Kim, Kim, Kim, Lee, Brandhuber, Brown, Pichini, Travaglini, Vives Matasan, Alessio, Gonzo, Shi,...
  - Special kinematic limit: soft, high energy, etc
    Di Vecchia, Heissenberg, Alessio, Rothstein, Saavedra, Del Duca, Gonzo, Rosi, ...
  - Trajectories: bound, large bend, etc
    Porto, Kälin, Liu, Adamo, Gonzo, Khalaf, Telem, Shen, De Angelis, Aoki, Cristofoli, ...
- Horizon absorption
  Goldberger, Rothstein, Aoude, Ochirov, Callum, Ruf, Gatica, Bautista, Huang, Kim, Chen, Hsieh, ...
- Connection with BHPT Bautista, Bonelli, Iossa, Tanzini, Zhou, Ivanov, Li, Parra-Martinez, ...
- Connection with self-force
  Cheung, Parra-Martinez, Rothstein, Shah, Wilson-Gerow, Kosmopolous, Solon, ...
- Dynamical Love number and BH analyticity Kosmopolous, Perrone, Solon, Correia, Gopalka, Isabella, Wolz, Caron-Huot, ... [see also Kobayashi's talk]
- Beyond perturbation theory Aoki, Cristofoli, Huang, ... [see also Jeong's talk]
- Quantum effect on curved background
  Aoude, O'Connell, Sergola, White, Aoki, Cristofoli, Jeong, Yoshimura [see also Yoshimura's talk]
- Beyond GR Alviani, Falkowski, Marinellis, Wilson-Gerow, ... [see also Marinellis' talk]

Conclusion and outlook

- EFT and on-shell methods are powerful tools for solving GR two-body problems
  - Lead to results that are otherwise hard to get
  - Can be applied to a large variety of problems
- Efficient computational tools propel the rapid progress of our field
  - On-shell methods for constructing integrands
  - IBP and DE for Feynman integrals and their generalization
- On-shell thinking provides a new perspective for solving an old problem: We welcome new ideas and challenges from different communities!

PHYSICAL REVIEW D **97**, 044038 (2018)

High-energy gravitational scattering and the general relativistic two-body problem

Thibault Damour*
Institut des Hautes Etudes Scientifiques, 35 route de Chartres, 91440 Bures-sur-Yvette, France

(Received 29 October 2017; published 26 February 2018)

A technique for translating the classical scattering function of two gravitationally interacting bodies into a corresponding (effective one-body) Hamiltonian description has been recently introduced [Phys. Rev. D **94**, 104015 (2016)]. Using this technique, we derive, for the first time, to second-order in Newton's constant (i.e. one classical loop) the Hamiltonian of two point masses having an arbitrary (possibly relativistic) relative velocity. The resulting (second post-Minkowskian) Hamiltonian is found to have a tame high-energy structure which we relate both to gravitational self-force studies of large mass-ratio binary systems, and to the ultra high-energy quantum scattering results of Amati, Ciafaloni and Veneziano. We derive several consequences of our second post-Minkowskian Hamiltonian: (i) the need to use special phase-space gauges to get a tame high-energy limit; and (ii) predictions about a (rest-mass independent) linear Regge trajectory behavior of high-angular-momenta, high-energy circular orbits. Ways of testing these predictions by dedicated numerical simulations are indicated. We finally indicate a way to connect our classical results to the quantum gravitational scattering amplitude of two particles, and we urge amplitude experts to use their novel techniques to compute the two-loop scattering amplitude of scalar masses, from which one could deduce the third post-Minkowskian effective one-body Hamiltonian.

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Thanks for listening!